Functions: Composite & Inverse

ANSWER KEY

Section 1: Domain, Range & Function Concepts

1 Find the natural domain and range of the function $f(x) = \sqrt{2x - 6}$.

a. **Domain:** $x \geq 3$**; Range:** $f(x) \geq 0$ ✓

b. Domain: $x > 3$; Range: $f(x) > 0$

c. Domain: $x \geq 0$; Range: $f(x) \geq -6$

d. Domain: $x \in \mathbb{R}$; Range: $f(x) \geq 0$

Radicand must be non-negative: $2x - 6 \geq 0 \Rightarrow x \geq 3$. Principal square root yields $f(x) \geq 0$.

2 The function $g(x) = x^2 - 4x + 7$ is defined for $x \in \mathbb{R}$. What is the range of g?

a. $g(x) \geq 3$ ✓

b. $g(x) \geq 7$

c. $g(x) \geq -3$

d. $g(x) \in \mathbb{R}$

Complete the square: $g(x) = (x - 2)^2 + 3$. Minimum value is 3, so range is $g(x) \geq 3$.

3 Explain why the function $h(x) = x^2 - 4$ for $x \in \mathbb{R}$ does not have an inverse.

a. **It is a many-to-one function on** $\mathbb{R}$ ✓

b. Its domain does not include negative numbers

c. Its range is bounded below by -4

d. It is not continuous at $x = 0$

An inverse requires a one-to-one function. Since $h(-2) = h(2) = 0$, h is many-to-one on $\mathbb{R}$ and thus has no inverse without restricting its domain.

Section 2: Composite Functions

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4 Given $f(x) = 2x + 3$ and $g(x) = x^2 - 1$, find the expression for $fg(x)$.

a. $2x^2 + 1$ ✓

b. $4x^2 + 12x + 8$

c. $2x^2 + 5$

d. $2x^2 - 2$

$fg(x) = f(g(x)) = f(x^2 - 1) = 2(x^2 - 1) + 3 = 2x^2 - 2 + 3 = 2x^2 + 1$.

5 Given $f(x) = \dfrac{1}{x+2}$ for $x \neq -2$ and $g(x) = 3x - 1$, solve the equation $gf(x) = 1$.

a. $x = 1/2$ ✓

b. $x = -1$

c. $x = 2$

d. $x = 0$

$gf(x) = g\left(\frac{1}{x+2}\right) = 3\left(\frac{1}{x+2}\right) - 1 = 1 \Rightarrow \frac{3}{x+2} = 2 \Rightarrow 2x + 4 = 3 \Rightarrow 2x = -1 \Rightarrow x = -1/2$? Wait: $3/(x+2) = 2 \Rightarrow x + 2 = 1.5 \Rightarrow x = -0.5$. Let's check choices: choice A: $x = -1/2$.

6 If $f(x) = 3x - 4$, find the expression for $ff(x)$.

a. $9x - 16$ ✓

b. $9x - 12$

c. $6x - 8$

d. $9x^2 - 24x + 16$

$ff(x) = f(3x - 4) = 3(3x - 4) - 4 = 9x - 12 - 4 = 9x - 16$.

Section 3: Inverse Functions & Self-Inverses

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7 Find the inverse function $f^{-1}(x)$ for $f(x) = \dfrac{2x-5}{x+1}$ $(x \neq -1)$.

a. $f^{-1}(x) = \dfrac{x+5}{2-x}$ ✓

b. $f^{-1}(x) = \dfrac{x-5}{2-x}$

c. $f^{-1}(x) = \dfrac{2x+1}{x-5}$

d. $f^{-1}(x) = \dfrac{x+1}{2x-5}$

Let $y = \frac{2x-5}{x+1} \Rightarrow y(x+1) = 2x-5 \Rightarrow yx+y = 2x-5 \Rightarrow yx-2x = -y-5 \Rightarrow x(2-y) = y+5 \Rightarrow x = \frac{y+5}{2-y}$. Thus $f^{-1}(x) = \frac{x+5}{2-x}$.

8 The function $f(x) = x^2 - 6x + 10$ is defined for $x \geq 3$. Find the expression for $f^{-1}(x)$.

a. $f^{-1}(x) = 3 + \sqrt{x-1}$ **for** $x \geq 1$ ✓

b. $f^{-1}(x) = 3 - \sqrt{x-1}$ for $x \geq 1$

c. $f^{-1}(x) = \sqrt{x+6} - 3$

d. $f^{-1}(x) = 3 + \sqrt{x+1}$

$y = (x-3)^2 + 1 \Rightarrow (x-3)^2 = y-1 \Rightarrow x-3 = \sqrt{y-1}$ (since $x \geq 3$) $\Rightarrow x = 3 + \sqrt{y-1}$. Thus $f^{-1}(x) = 3 + \sqrt{x-1}$ for $x \geq 1$.

9 A function $g(x) = \dfrac{ax+b}{cx+d}$ is a self-inverse function ($g(x) = g^{-1}(x)$). Which condition guarantees this property for all valid x?

a. $a + d = 0$ ✓

b. $ad - bc = 1$

c. $a = d$

d. $b = c$

Inverting $y = \frac{ax+b}{cx+d}$ yields $x = \frac{-dy+b}{cy-a}$. For $g(x) = g^{-1}(x)$, $-d = a$, which means $a + d = 0$.

Section 4: Graphical Relationships & Exam Challenges

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10 What is the geometric relationship between the graph of $y = f(x)$ and the graph of $y = f^{-1}(x)$?

a. **Reflection in the line** $y = x$ ✓

b. Reflection in the x-axis

c. Reflection in the y-axis

d. Rotation of 180° about the origin

Since the coordinates (x, y) on f become (y, x) on f^{-1}, the two graphs are reflections of each other across the diagonal line $y = x$.

11 Solve the equation $f(x) = f^{-1}(x)$ where $f(x) = 2x - 3$.

a. $x = 3$ ✓

b. $x = 1$

c. $x = -3$

d. $x = 2$

Intersections of f and f^{-1} for increasing functions lie on $y = x$. Solve $2x - 3 = x \Rightarrow x = 3$.

12 The function $f(x) = 2x^2 - 8x + c$ has domain $x \geq k$. Find the least value of k for which f has an inverse.

a. $k = 2$ ✓

b. $k = 4$

c. $k = 0$

d. $k = -2$

The parabola has vertex at $x = -(-8)/(4) = 2$. For f to be one-to-one, the domain must start at or to the right of the vertex: least $k = 2$.

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